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 Oefeningenreeks 2B nr. 11

$$f(x) = \tan(\operatorname{Bgsin}(\cos(\operatorname{Bgtan}(\cos(\operatorname{Bgtan}\sqrt{3}))))))$$

$$\operatorname{Bgtan}\sqrt{3} = \frac{\pi}{3} \text{ want } \tan\frac{\pi}{3} = \sqrt{3}$$

$$\text{neem } y = \operatorname{Bgtan}\sqrt{3} \text{ dan is } \cos y = \cos\frac{\pi}{3} = \frac{1}{2}$$

$$\text{neem } z = \cos y \text{ dan is } \operatorname{Bgtan} z = \operatorname{Bgtan}\frac{1}{2}$$

$$\text{neem } u = \operatorname{Bgtan}\frac{1}{2} \text{ dan is } \tan u = \frac{1}{2}$$

we zoeken nu wat $\cos u$ aan gelijk is.

We kunnen hiervoor de identiteit $\cos^2 u = \frac{1}{1 + \tan^2 u}$ gebruiken:

$$\cos^2 u = \frac{1}{1 + \tan^2 u}$$

$$\cos u = \sqrt{\frac{1}{1 + \tan^2 u}}$$

$$= \sqrt{\frac{1}{1 + \left(\frac{1}{2}\right)^2}}$$

$$= \sqrt{\frac{1}{\frac{5}{4}}}$$

$$= \sqrt{\frac{4}{5}}$$

$$= \frac{\sqrt{4}}{\sqrt{5}}$$

$$= \frac{2}{\sqrt{5}} \text{ aangezien } u \in \left] \frac{-\pi}{2}, \frac{\pi}{2} \right[\text{ is de cosinus positief.}$$

$$\text{Neem } v = \operatorname{Bgsin}\frac{2}{\sqrt{5}} \text{ dan is } \sin v = \frac{2}{\sqrt{5}}$$

we zoeken nu wat $\tan v$ aan gelijk is:

$$\tan v = \frac{\sin v}{\cos v}$$

hiervoor moeten we nog de cosinus kennen.

We kunnen de identiteit $\sin^2 v + \cos^2 v = 1$ gebruiken:

$$\sin^2 v + \cos^2 v = 1$$

$$\cos^2 v = 1 - \sin^2 v$$

$$\cos^2 v = 1 - \left(\frac{2}{\sqrt{5}} \right)^2$$

$$= 1 - \frac{4}{5}$$

$$= \frac{1}{5}$$

$$\cos v = \sqrt{\frac{1}{5}}$$

$$= \frac{1}{\sqrt{5}} \text{ aangezien } v \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right] \text{ is de cosinus positief.}$$

Nu we de sinus en cosinus hebben gevonden kunnen we de tangens berekenen:

$$\tan v = \frac{\frac{2}{\sqrt{5}}}{\frac{1}{\sqrt{5}}}$$

$$= \frac{2}{\sqrt{5}} \cdot \frac{\sqrt{5}}{1}$$

$$= \frac{2\sqrt{5}}{1\sqrt{5}}$$

$$= \frac{2}{1}$$

$$= 2$$

$$\text{dus is } \tan(\operatorname{Bgsin}(\cos(\operatorname{Bgtan}(\cos(\operatorname{Bgtan} \sqrt{3})))))) = 2$$

References